

STAT 516 - Homework 4

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(9 points)** A discrete random variable X has probability mass function

$$P(X = x) = \begin{cases} p, & x = 0, \\ 2p, & x = 2, \\ 3p, & x = 4, \\ 1 - 6p, & x = 6. \end{cases}$$

You are given that $E[X] = 3.6$. Determine p .

2. **(9 points)** A discrete random variable X takes values 1, 2, 4, 5 with probabilities

$$P(X = 1) = 0.10, \quad P(X = 2) = 0.25, \quad P(X = 4) = 0.40, \quad P(X = 5) = 0.25.$$

Calculate $E[X]$ and $\text{Var}(X)$.

3. **(10 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} cx^2, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find c , and then calculate $E[X]$.

4. **(9 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{2|x|}{13}, & -2 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E[X]$.

5. **(9 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{3}{26}x^2, & 1 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E\left[\frac{1}{X}\right]$.

6. **(9 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \lambda^2 x + \frac{3}{2}\lambda^3 x^2, & 0 < x < \frac{1}{\lambda}, \\ 0, & \text{otherwise,} \end{cases}$$

where $\lambda > 0$. Verify that this is a valid density function.

7. **(10 points)** Let X be a continuous random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 1, \\ \frac{(x-1)^2}{9}, & 1 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

Calculate $P(X < 3)$ and find the density function $f(x)$.

8. **(10 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{1}{3}e^{-x/3}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E[X]$.

9. **(9 points)** You are given

$$E[X] = 6, \quad \text{Var}(X) = 16.$$

Calculate

$$E[(2X - 3)^2].$$

10. **(9 points)** Let X be a random variable such that

$$E[X] = 3, \quad E[X^3] = 36, \quad E[(X - 3)^3] = 0.$$

Calculate $\text{Var}(X)$.

11. **(7 points)** The annual repair cost X for a machine has mean 500 dollars and variance 900. After a price increase, the new repair cost is

$$Y = 1.15X + 20.$$

Calculate $\text{Var}(Y)$.